

- Performance enhancements in the guest SMP code
- Better support for huge pages
- Better emulation accuracy
- Support for MSI
- Fixes and improvements in the device assignment code

What this means is that guests will be more responsive, and there's more efficient interrupt handling for devices that are assigned from the host (PCI passthrough), as well as better stability.

For developers, there's improved guest debugging support and nested virtualisation on AMD hosts, which allows one to run a KVM guest within a KVM guest.

Changes in the generic kernel code that are useful for virtualisation include:

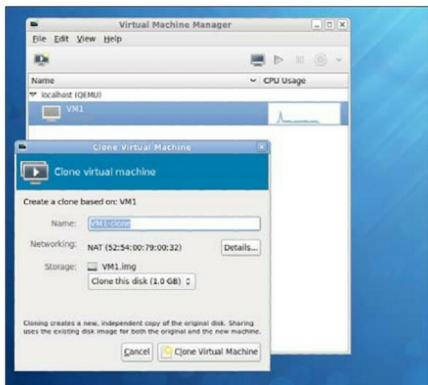
- SR-IOV (Single Root I/O Virtualisation) support
- KSM (Kernel SamePage Merging)
- VirtIO performance enhancements and fixes: block and net devices are much faster and more stable

SR-IOV support enables sharing of a physical PCI device among multiple guests. The performance while accessing such devices from the guests is at par with the performance of using the device from the host itself. This feature needs supported hardware. As of 2.6.31, the Intel 82576 Gigabit Ethernet Controller and the NetXen X3100 cards are supported. This feature needs driver support, both in the host as well as the guest. This feature builds on top of the device assignment code that was introduced in Fedora 11.

The KSM feature enables sharing of identical RAM contents to reduce physical memory usage by a host, improving the guest density. This is discussed in greater detail in a separate article titled "Introducing Kernel SamePage Merging" in this issue.

The device model (the virtual computer) that is presented to the guests is instantiated by QEMU. A lot of user-facing changes happen here. A short list of notable features that were introduced include:

- Installs guests from CD/DVD



Virt manager in action

images located remotely over the HTTP protocol instead of having to download the images first.

- Shows the host CPU type as the guest CPU. This results in better utilisation of the host CPU features by guest applications, speeding up some of them that rely on newer features. However, enabling this feature may come at the cost of guest migration.
- Allows for the setting of migration speed and the setting of a maximum allowed downtime while live-migrating guests.
- Migration code has got a lot of fixes, which means more stable guests after live migrations.
- Many stabilisation fixes in the block code.
- The qcow2 file format saw a lot of performance and stabilisation fixes.
- Fixes and performance improvements in the virtio-block and virtio-net code.
- The ability to hot plug NICs in guests.
- A stable guest ABI, which ensures the hardware that the guest sees remains the same across

QEMU upgrades. This is helpful especially for Windows guests, which may need reactivation on hardware changes.

- MSI support: the guests can now use MSI/MSI-X, resulting in better performance.
- Support for debugging guests.

For developers, there's a lot of good news: a majority of the QEMU code was overhauled and it's in much better shape than earlier. It's now much easier to catch bugs at code review time. The development has also moved to a git tree from svn.

The libvirt development continued at a fast pace as well, with the addition of the following features, among others:

- Present a stable guest ABI
- NIC hot plug for guests
- Improved security by starting guests with lower privileges. Allows KVM guests to be started by unprivileged users.
- Ability to run guests with hugepage support: this helps reduce the number of memory pages in the host. This results in better performance by reduced